

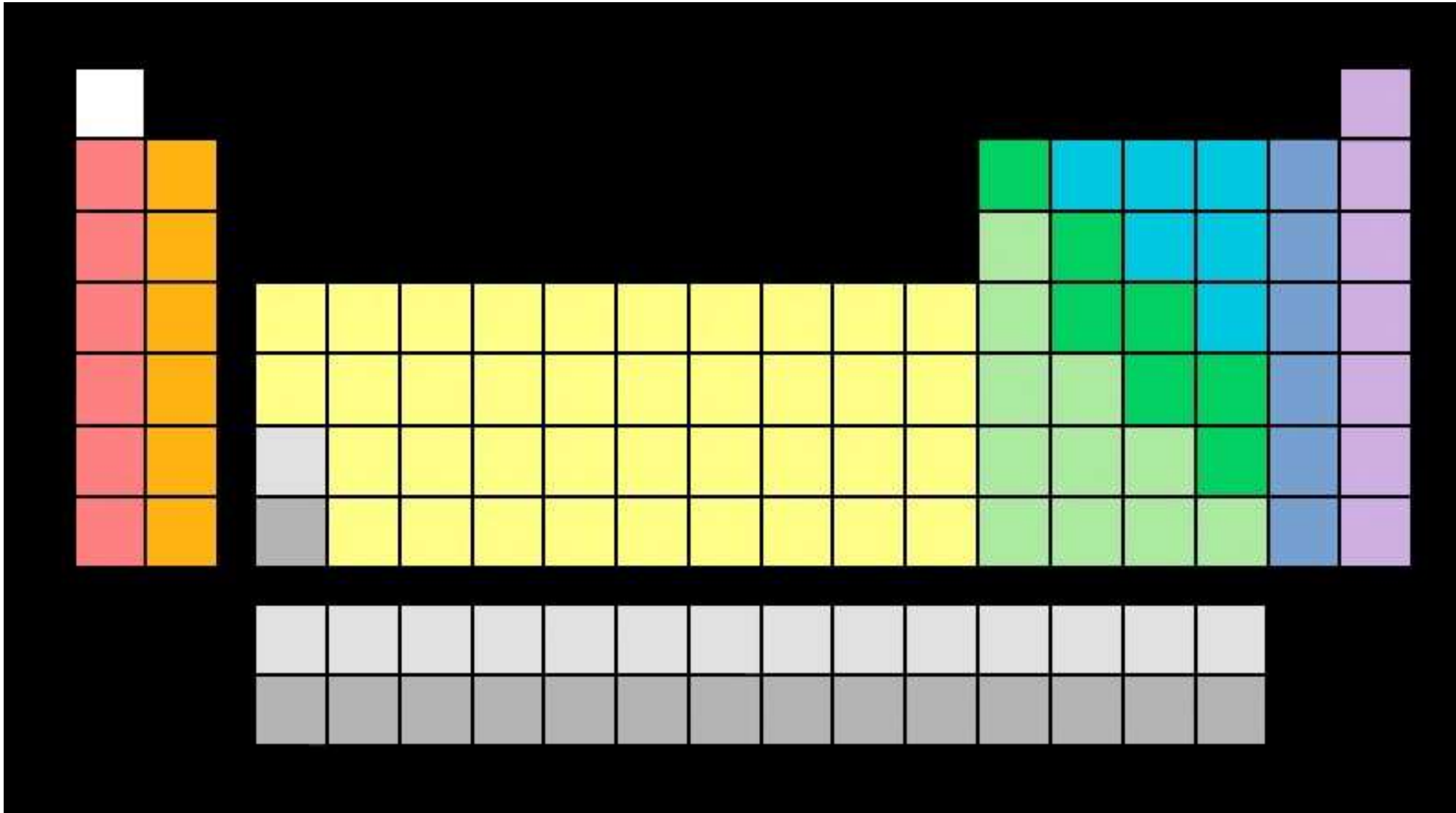
CHM101

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Chemistry for Life®

PERIODIC TABLE OF ELEMENTS

GROUP																18			
1																2			
1	H Hydrogen 1.008															He Helium 4.003			
2	Li Lithium 6.94	Be Beryllium 9.012																	
3	Na Sodium 22.99	Mg Magnesium 24.31																	
4	K Potassium 39.1	Ca Calcium 40.08	Sc Scandium 44.96	Ti Titanium 47.88	V Vanadium 50.94	Cr Chromium 52.00	Mn Manganese 54.94	Fe Iron 55.85	Co Cobalt 58.93	Ni Nickel 58.69	Cu Copper 63.55	Zn Zinc 65.39	Ga Gallium 69.72	Ge Germanium 72.64	As Arsenic 74.92	Se Selenium 78.96	Br Bromine 79.90	Kr Krypton 83.80	
5	Rb Rubidium 85.47	Sr Strontium 87.62	Y Yttrium 88.91	Zr Zirconium 91.22	Nb Niobium 92.91	Mo Molybdenum 95.94	Tc Technetium (98)	Ru Ruthenium 101.1	Rh Rhodium 102.9	Pd Palladium 106.4	Ag Silver 107.9	Cd Cadmium 112.4	In Indium 114.8	Sn Tin 118.7	Sb Antimony 121.8	Te Tellurium 127.6	I Iodine 126.9	Xe Xenon 131.3	
6	Cs Cesium 132.9	Ba Barium 137.3	57-71 Lanthanides		Hf Hafnium 178.5	Ta Tantalum 180.9	W Tungsten 183.8	Re Rhenium 186.2	Os Osmium 190.2	Ir Iridium 192.2	Pt Platinum 195.1	Au Gold 197.0	Hg Mercury 200.6	Tl Thallium 204.4	Pb Lead 207.2	Bi Bismuth 209.0	Po Polonium (209)	At Astatine (210)	Rn Radon (222)
7	Fr Francium (223)	Ra Radium (226)	89-103 Actinides		Rf Rutherfordium (261)	Db Dubnium (262)	Sg Seaborgium (266)	Bh Bohrium (264)	Hs Hassium (277)	Mt Meitnerium (268)	Ds Darmstadtium (281)	Rg Roentgenium (282)	Cn Copernicium (285)	Nh Nihonium (286)	Fl Flerovium (289)	Mc Moscovium (290)	Lv Livermorium (293)	Ts Tennessine (294)	Og Oganesson (294)
Lanthanides		La Lanthanum 138.9	Ce Cerium 140.1	Pr Praseodymium 140.9	Nd Neodymium 144.2	Pm Promethium (145)	Sm Samarium 150.4	Eu Europium 152.0	Gd Gadolinium 157.3	Tb Terbium 158.9	Dy Dysprosium 162.5	Ho Holmium 164.9	Er Erbium 167.3	Tm Thulium 168.9	Yb Ytterbium 173.0	Lu Lutetium 174.9			
Actinides		Ac Actinium (227)	Th Thorium 232.0	Pa Protactinium 231.0	U Uranium 238.0	Np Neptunium 237.0	Pu Plutonium 244.0	Am Americium 243.0	Cm Curium 247.0	Bk Berkelium 247.0	Cf Californium 251.0	Es Einsteinium 252.0	Fm Fermium 257.0	Md Mendelevium 258.1	No Nobelium 259.1	Lr Lawrencium 262.1			

78

Pt

Platinum

195.1

Atomic Number

Symbol

Name

Average Atomic Mass

- Alkali Metals
- Alkaline Earth Metals
- Transition Metals
- Other Metals
- Metalloids
- Non-metals
- Halogens
- Noble Gases
- Lanthanides
- Actinides

78
Pt
Platinum
195.1

Atomic Number

Symbol

Name

Average Atomic Mass

Effective Nuclear Charge

Slater's Rule

- **For an electron in *ns* and *np* orbital:**

1. Write the electronic configuration and group them accordingly:

$(1s)(2s, 2p)(3s, 3p)(3d)(4s, 4p)(4d)(4f)(5s, 5p), \dots$

2. Electrons in any group to the right of the electron of interest i.e. of the (ns, np) group do not contribute to the shielding constant.

3. All of the other electrons in the (ns, np) group, shield 'the electron' to an extent of 0.35 each

4. All the electrons in the $(n-1)$ shell shield to an extent of 0.85 each

5. All the electrons in the $(n-2)$ or lower shell shield completely; that is their contribution is 1.0 each

- **When the electron is in *nd* or *nf* orbital:**

- Rules 1,2 and 3 are same as the previous one but rules 4 and 5 become-
- All electrons in groups lying to the left of the *nd* or *nf* group contribute 1.00 towards the shielding constant.

$$Z^* = Z - \sigma$$

Z^* = Effective nuclear charge

Z = Nuclear charge

σ = Shielding constant

Practice Problem

1. Using Slater's rules, calculate Z^* for the following electrons:
a 3p electron in P

Practice Problem

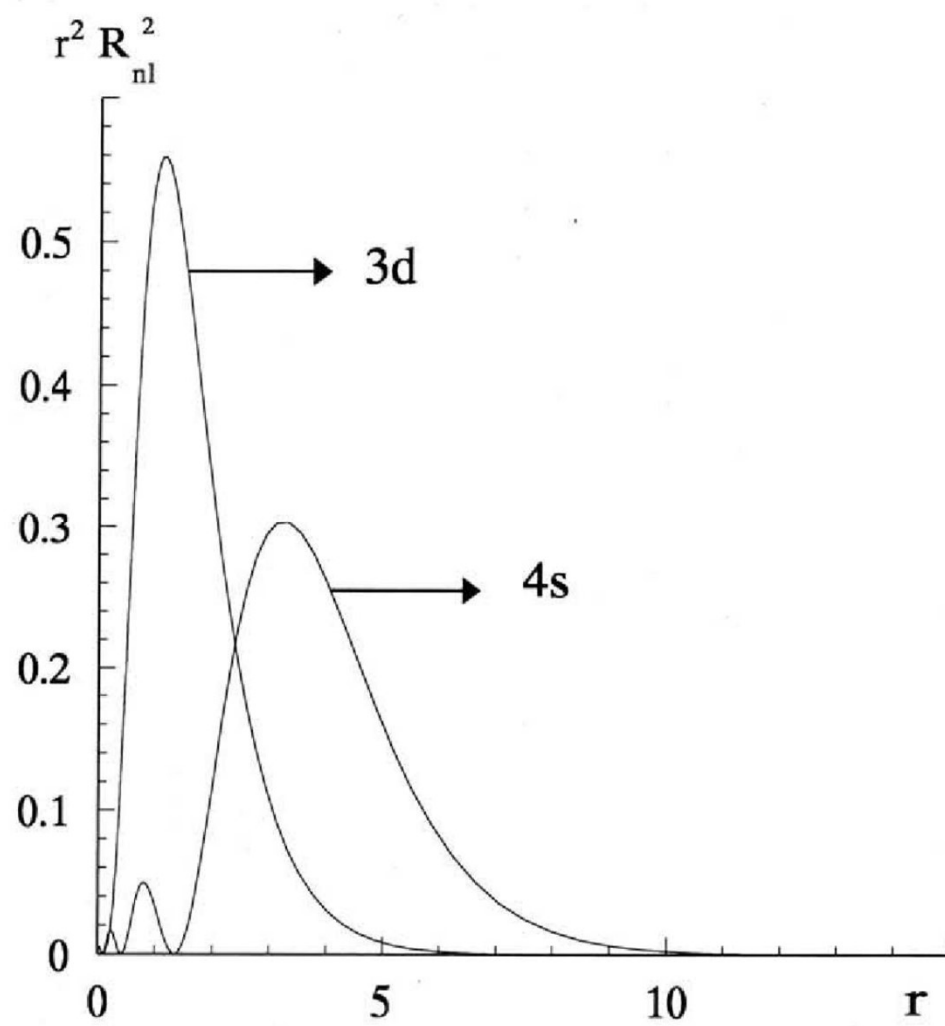
2. Using Slater's rules, calculate Z^* for the following electrons:

a 3d electron in Mn

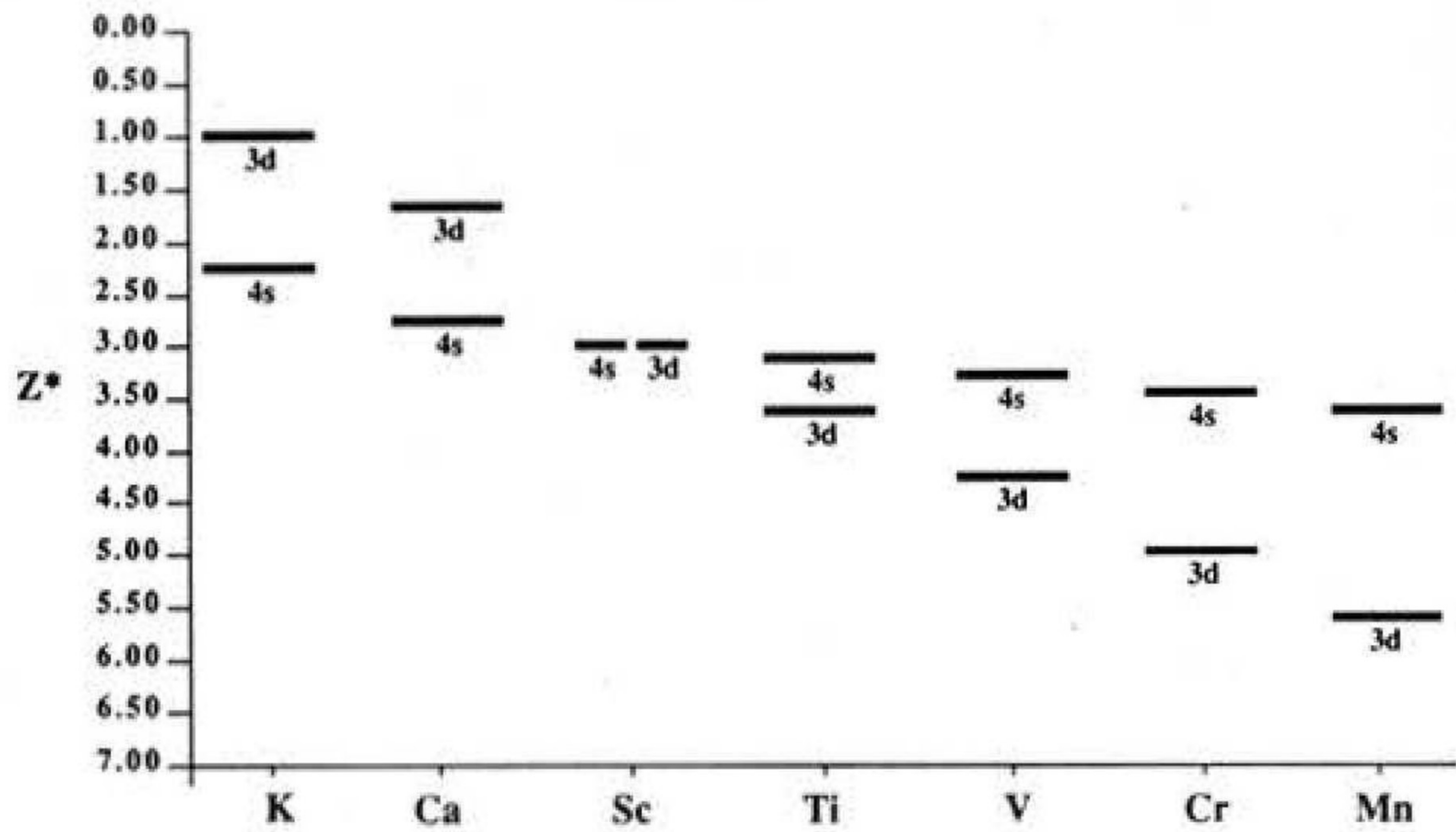
A 4s electron in Mn

Relative Energies of ns and $(n-1)d$ orbitals

- ns orbital fill before the $(n-1)d$ orbitals
- While for ionization process, electrons are removed from the ns orbitals first

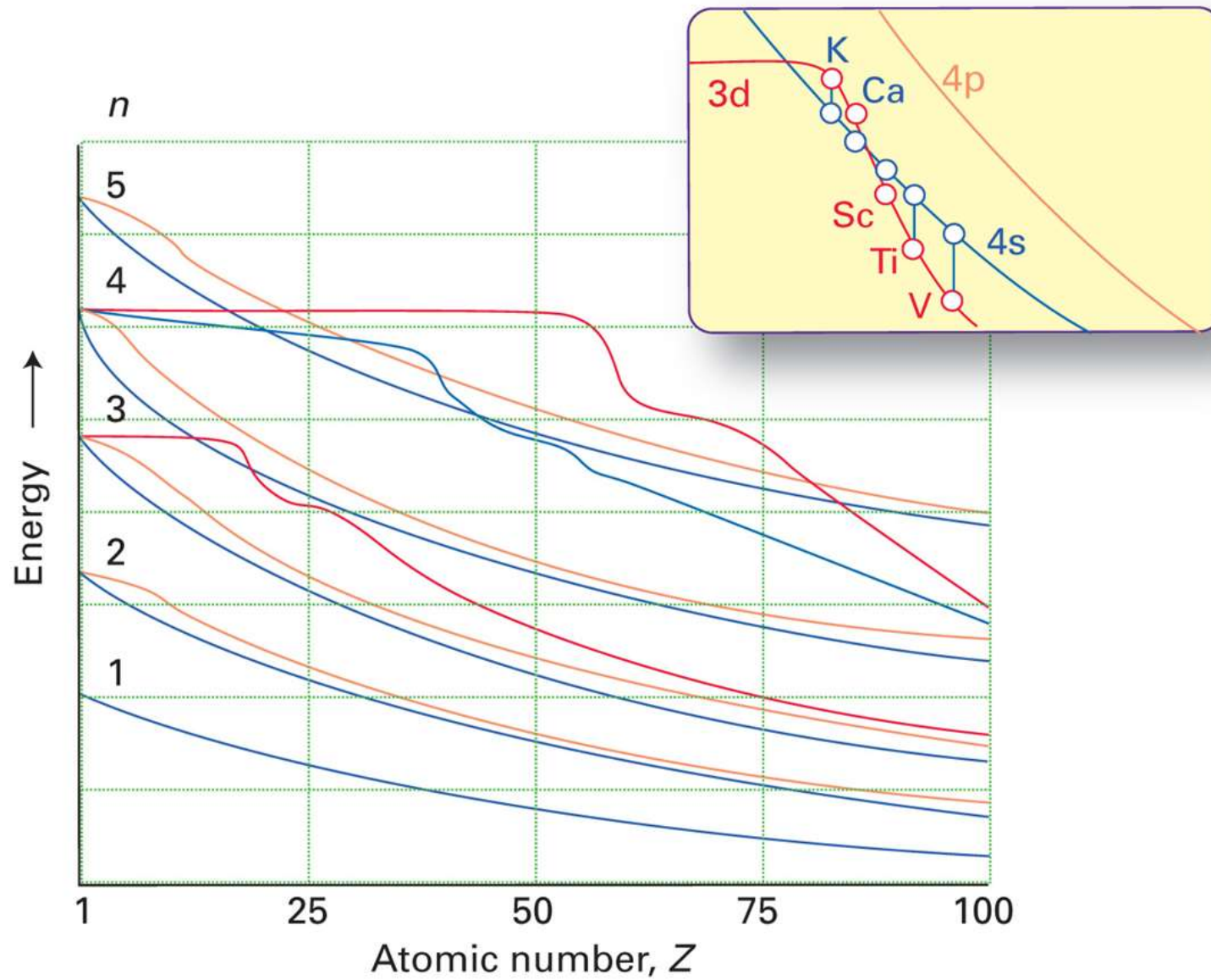


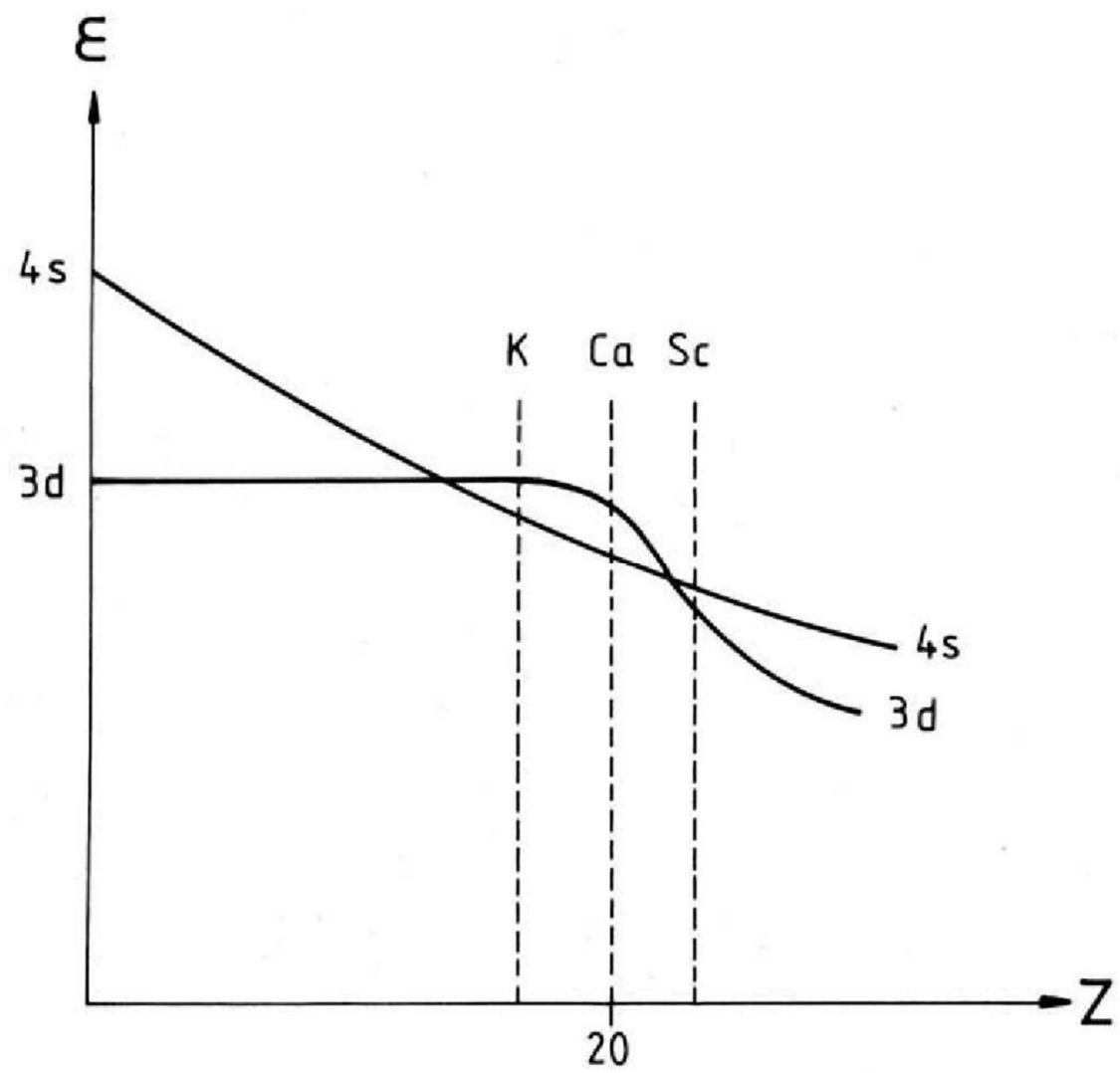
- Using Slater's rule find the correct electronic configuration of K
- Calculate the Z^* for a 3d electron and a 4s electron of Ti



Calculated Effective Nuclear Charge (Z^*) Values for K through Mn

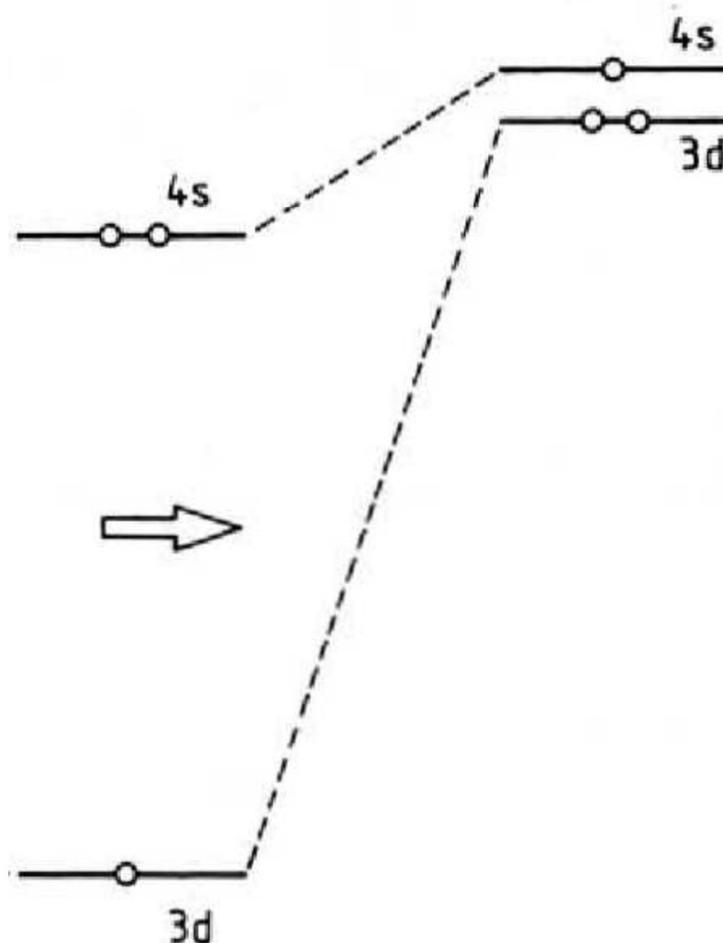
Element	Z^* for 4s electron	Z^* for 3d electron
K	2.20	1.00
Ca	2.85	1.65
Sc	3.00	3.00
Ti	3.15	3.65
V	3.30	4.30
Cr	3.45	4.95
Mn	3.60	5.60



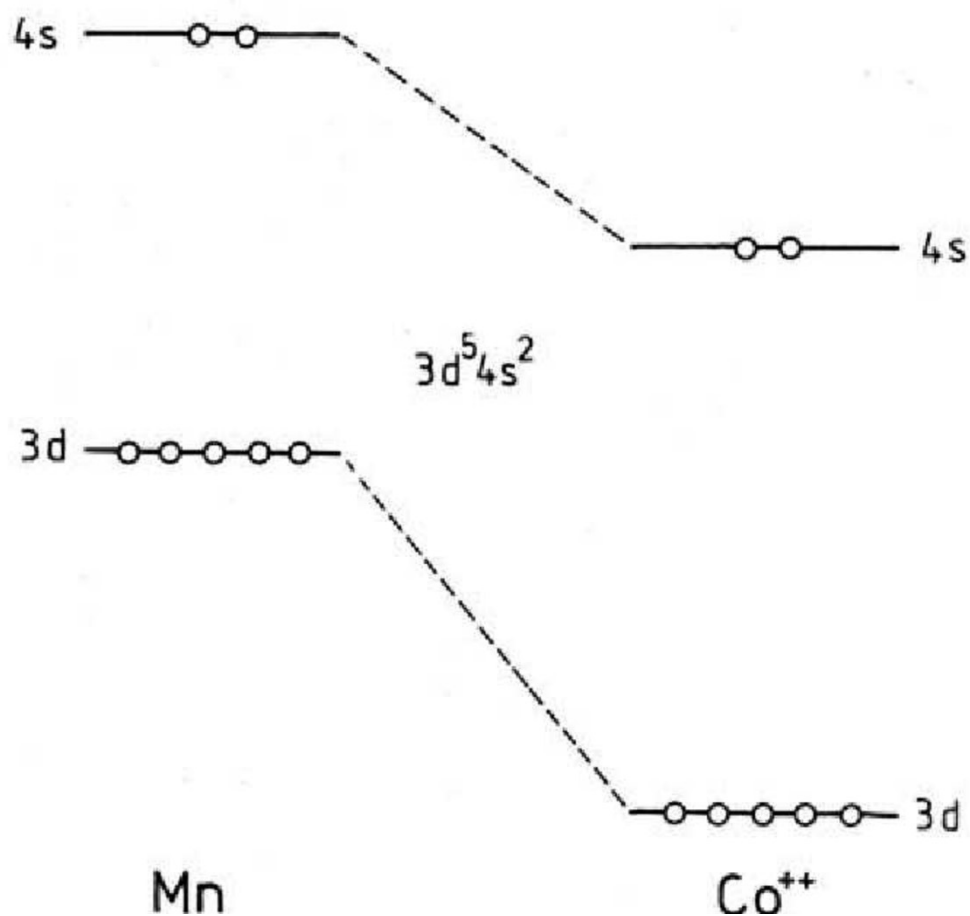


$$\epsilon_{3d}(3d^n 4s^2) < \epsilon_{3d}(3d^{n+1} 4s^1) < \epsilon_{3d}(3d^{n+2} 4s^0)$$

$$\epsilon_{4s}(3d^n 4s^2) < \epsilon_{4s}(3d^{n+1} 4s^1)$$



Orbital Energies in Transition Metal Ions



A

